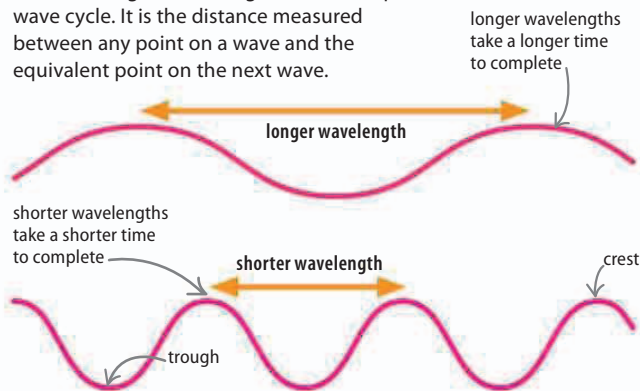


Measuring waves

Waves have three important measurements. These are wavelength, frequency, and amplitude.

▽ Wavelength

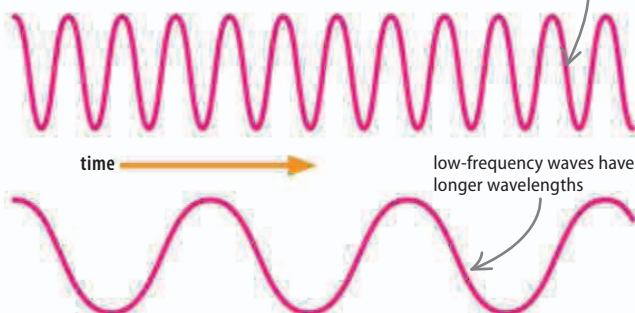
The wavelength is the length of one complete wave cycle. It is the distance measured between any point on a wave and the equivalent point on the next wave.



▽ Frequency

This is the number of waves passing any point in one second. The unit of frequency is the hertz (Hz).

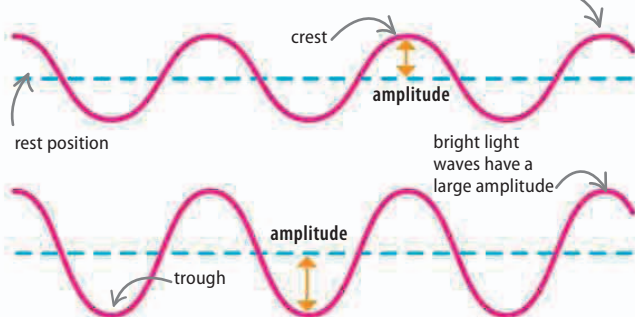
waves with a short wavelength have a higher frequency than those with longer wavelengths



▽ Amplitude

The amplitude is the height of a crest or trough as the wave travels, measured from the central rest position.

dim light waves have a small amplitude



Wave speed

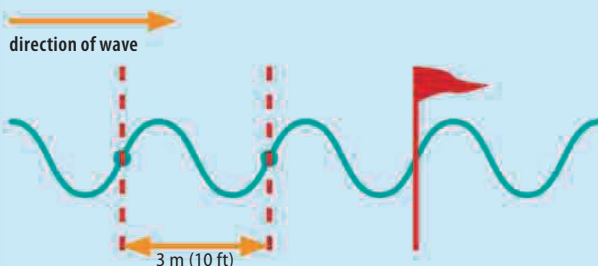
The speed of a wave is related to its frequency and wavelength. They are linked by this equation:

wave speed (meters per second) wavelength (meters) frequency is the number of waves per second (hertz (Hz))

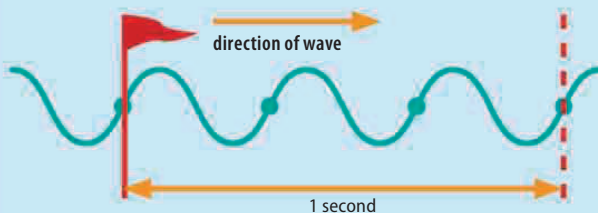
$$\text{Wave speed} = \text{Wavelength} \times \text{Frequency}$$

Calculating wave speed

The waves below are traveling across water. To find the wave speed, multiply the wavelength by the frequency.



△ The wavelength has been worked out—each wave is 3 m (10 ft) long. A marker is used to help count the number of waves in one second.



△ One second later Three waves have passed the marker, so the frequency is 3 Hz.

wavelength frequency wave speed

$$3 \text{ m} \times 3 \text{ Hz} = 9 \text{ m/s}$$